



Platinum single-atom catalysts anchored on van der Waals heterostructure support for durable hydrogen evolution

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ABSTRACT The activity and stability of single-atom catalysts (SACs) are intimately associated with the structure of supports. Herein, by employing a van der Waals (vdW) heterostructure support, we construct a highly active and durable Pt SAC for hydrogen evolution reaction (HER). The unique support consists of monolayer MoS₂ attaching on hierarchical N-doped carbon nanocages (hNCNC), on which Pt presents as individual single atoms on the hNCNC and as island-like single-atom layers on the MoS₂. The optimized Pt₁-MoS₂/hNCNC demonstrates low overpotential (11 mV at 10 mA cm⁻²) and high mass activity (5.6 A mg_{Pt}⁻¹ at -20 mV) in 0.5 M H₂SO₄ solution, outperforming commercial Pt/C. Impressively, the Pt₁-MoS₂/hNCNC exhibits improved long-term stability in proton exchange membrane water electrolyzer relative to commercial Pt/C. The excellent HER performance is attributed to the regulated electronic structure, robust interaction of Pt atoms with MoS₂/hNCNC and facilitated charge transfer. This study establishes an innovative strategy to develop a highly active and durable Pt SAC using vdW heterostructure supports.

Keywords: hydrogen evolution reaction, van der Waals heterostructures, single-atom catalysts, platinum, proton exchange membrane water electrolyzer

INTRODUCTION

Hydrogen is an appealing energy alternative to fossil fuels owing to its high efficiency, zero emissions, and abundant availability [1]. Among various hydrogen production technologies, proton exchange membrane water electrolysis (PEMWE) powered by renewable energy represents one of the most feasible techniques for commercial use because of its advantages of high power-to-gas conversion efficiency, compact structure and fast response [2,3]. However, PEMWE relies on noble metal catalysts; for instance, the cathode catalyst for hydrogen evolution reaction (HER) is Pt/C with typical areal loading of 0.4–0.5 mg_{Pt} cm⁻². The high cost and scarcity of Pt pose major obstacles to the wide application of PEMWE [4,5]. In the past decade, single-atom catalysts (SACs) have garnered significant interest as a promising breakthrough for reducing the usage of noble metals [6,7]. The individual metal atoms are often anchored on an appro-

priate support such as nanocarbons [4,8], metal compounds [9,10], and MXenes [11], not only to minimize surface energy but also to fine-tune the electronic structure and catalytic activity thereof. Hence, the activity and stability of SACs are intimately associated with the structure of supports [12–15]. Nanocarbon-supported Pt SACs have shown excellent HER performance due to their high conductivity, abundant morphology and electronic tunability [13]. Recently, by using hierarchical N-doped carbon nanocages (hNCNC) as support, we have developed a Pt₁/hNCNC catalyst through synergic micropore trapping and N anchoring, and the resulting active sites with Pt-N₂ configuration have demonstrated ultralow HER overpotential [8,14,15]. The PEMWE using Pt₁/hNCNC cathode exhibits quite good stability with a small voltage increase of 9 mV in 100 h operation [14]. To further bolster the HER stability of the Pt SACs, innovative construction strategies need to be developed.

MoS₂, as a typical two-dimensional (2D) material, has long been recognized as an efficient catalyst for HER. MoS₂ can also act as advanced support to anchor Pt atoms atop the Mo atoms, exhibiting much enhanced HER performance due to the near-zero H adsorption free energy (ΔG_{H^*}) and strong Pt-support interaction [16]. However, the MoS₂-based catalysts usually suffer from their poor conductivity and restacking issues [17,18]. Recently van der Waals (vdW) heterostructures made by restacking different 2D materials emerge as multifunctional materials [19–21]. The vdW heterostructure of MoS₂ and hNCNC can provide more anchoring sites for Pt atoms, faster charge/mass transport and regulated electronic structure, offering an intriguing strategy and new possibilities for boosting HER performance.

Herein, a MoS₂/hNCNC heterostructure support was simply prepared by pyrolyzing an hNCNC-ammonium thiomolybdate composite, in which MoS₂ monolayers adhere to the hNCNC surface through vdW interaction, and individual Pt atoms were immobilized on the MoS₂/hNCNC through a simple impregnation-drying process. The resulting Pt₁-MoS₂/hNCNC exhibits ultralow overpotential, high mass activity, and excellent stability in a 0.5 M H₂SO₄ solution, outperforming commercial Pt/C (20 wt%). The PEMWE using Pt₁-MoS₂/hNCNC as cathode catalyst with an ultralow Pt loading of 40 μ g_{Pt} cm⁻² demonstrates remarkable performance, achieving the current densities

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of 1.0 and 2.0 A cm⁻² at the cell voltages as low as 1.70 and 1.92 V, respectively. More impressively, the PEMWE presents improved stability in 500 h operation at a constant current density of 1 A cm⁻², with only a voltage increase of 40 μV h⁻¹, smaller than the Pt/C (44 μV h⁻¹) and Pt₁/hNCNC (90 μV h⁻¹) [14]. This study demonstrates an innovative strategy to develop high-activity and durable Pt SACs using vdW heterostructure supports.

EXPERIMENTAL SECTION

Synthesis

The hNCNC support was synthesized through an *in situ* MgO template method at 800 °C with pyridine precursor [22–24]. The MoS₂/hNCNC vdW heterostructure support was prepared by the pyrolysis of hNCNC-supported ammonium tetrathiomolybdate, (NH₄)₂MoS₄. Typically, an appropriate amount of (NH₄)₂MoS₄ was dissolved in 20 mL ultrapure water, and 100 mg hNCNC was added into the solution, followed by the sonication treatment for 30 min and drying at 70 °C under stirring. The resulting powder was then heated to 800 °C under a 10%-H₂/Ar atmosphere for 2 h, leading to the MoS₂/hNCNC. The Pt₁-MoS₂/hNCNC catalyst was prepared by the impregnation-drying process [8,14,15]. Briefly, 50 mg MoS₂/hNCNC was dispersed in 80 mL of distilled water and heated to 70 °C, and then a specific volume of chloroplatinic acid solution (H₂PtCl₆, 2 mg mL⁻¹) was added dropwise. After stirring at 70 °C for 10 h, the suspension was filtered, washed thoroughly with water and ethanol, and dried overnight at 70 °C. By varying the amounts of (NH₄)₂MoS₄ and H₂PtCl₆, a series of Pt₁-MoS₂/hNCNC catalysts with different MoS₂ loadings (9.3, 20.8, 32.7, 38.7 wt%) and Pt loadings (0, 3.8, 6.1, 7.3 wt%) were prepared for optimization. For comparison, the Pt₁/hNCNC catalyst (without MoS₂) was prepared by the impregnation-drying process, with a Pt content of 3.8 wt%. The detailed information about the characterization methods is shown in Supplementary information.

Electrochemical measurements in a three-electrode system

The HER performance of the catalysts was evaluated in a three-electrode electrochemical cell on CHI 660C electrochemical workstation, with a graphite rod counter electrode and an Ag/AgCl (3 M KCl) reference electrode. The working electrode was prepared as follows: 2 mg catalyst was dispersed in a mixed solution of 800 μL of ultrapure water, 200 μL of ethanol, and 40 μL of Nafion (Dupont, 5 wt%). The mixture was then transferred to an ultrasonic bath to form a homogeneous catalyst ink. Fresh ink (10 μL) was dropped onto the glassy carbon zone (5 mm in diameter) of a rotating disk electrode (RDE, Pine Instrument Co.), and dried naturally at room temperature. The HER tests were conducted in N₂-bubbled 0.5 M H₂SO₄ solution with a rotation rate of 1600 r min⁻¹. All the potentials were converted to versus reversible hydrogen electrode (RHE), i.e., $E_{\text{RHE}} = E_{\text{Ag/AgCl}} + 0.209 + 0.059 \times \text{pH}$.

Electrochemical measurements in PEMWE

The measured electrode area of the PEMWE is 2 cm × 2 cm. The cathode ink was prepared by dispersing 15 mg Pt catalyst in a mixture of 0.9 g isopropyl alcohol, 0.9 g deionized water, and 100 mg Nafion solution (5 wt%), with an ionomer-to-catalyst ratio of 1:3. The anode ink was prepared by dispersing 60 mg

iridium oxide (IrO₂) in a mixture of 0.9 g isopropyl alcohol, 0.9 g deionized water, and 300 mg Nafion solution (5 wt%), with an ionomer-to-catalyst ratio of 1:4. The two inks were coated onto both sides of Nafion 115 through ultrasonic spraying to obtain a catalyst-coated membrane (CCM). The anode contains IrO₂ of 2.0 mg cm⁻², and the cathode contains Pt catalysts (20 wt% Pt/C or Pt₁-MoS₂/hNCNC) with a low Pt areal loading of 40 μg_{Pt} cm⁻². The membrane electrode array (MEA) was assembled by hot-pressing bare titanium sheets (0.4 mm thick) at 80 °C, 2 MPa for 1 min. The PEMWE was operated at 80 °C using deionized water as reactant with a flow rate of 20 mL min⁻¹. Before testing, the PEMWE was activated for 1 h at 1.0 A cm⁻². During the polarization test, the cell voltage was increased from 1.3 to 2.3 V in 0.01 V increments, holding each step for 10 s. The stability was evaluated over 500 h at 80 °C and 1.0 A cm⁻². All the PEMWE results were presented without *iR* compensation.

RESULTS AND DISCUSSION

The hNCNC support has multiscale pore structure, large specific surface area (1329 m² g⁻¹), high N content (7.42 at%) and bulk conductivity (218.6 S m⁻¹) (Fig. S1). Our previous studies have demonstrated that the hNCNC can provide abundant anchoring sites for isolated Pt atoms, and its multiscale pore structure, high conductivity and excellent wettability promote mass/charge synergistic transport, which benefits the HER performance of hNCNC-supported Pt SACs [8,14,15]. Four MoS₂/hNCNC composites were synthesized by changing the feeding amount of (NH₄)₂MoS₄, with the respective MoS₂ loading of 9.3, 20.8, 32.7 and 38.7 wt% (Fig. S2). Their bulk conductivity increases with decreasing the MoS₂ loading, being 178.7 and 169.4 S m⁻¹ for the 9.3 wt%- and 20.8 wt%-MoS₂/hNCNC, respectively (Table S1). Scanning electron microscopy (SEM) observation shows that the MoS₂/hNCNC support maintains the hierarchical porous structure of hNCNC (Fig. S3), favoring the mass transfer to the active sites [25]. Elemental mapping confirms the uniform distributions of C, N, and O in MoS₂/hNCNC, while S and Mo elements are concentrated in island-like patterns (Fig. S4), confirming the successful fabrication of MoS₂/hNCNC.

With these MoS₂/hNCNC supports, four Pt-MoS₂/hNCNC catalysts with a fixed Pt loading (~3.8 wt%) were constructed. High-resolution transmission electron microscopy (HRTEM) characterizations indicate that the MoS₂ layers adhere to the outer surface of hNCNC, and no Pt species are observed (Fig. S5). The layer number of MoS₂ increases with increasing the MoS₂ loading; that is, the MoS₂ exists in a monolayer structure for the low-loading MoS₂/hNCNC (with 9.3 and 20.8 wt% of MoS₂), whereas it is present in both monolayer and multilayer structures in high-loading MoS₂/hNCNC (with 32.7 and 38.7 wt% of MoS₂). X-ray diffraction (XRD) analysis indicates no diffraction peaks for Pt and MoS₂ are present in the low-loading MoS₂/hNCNC, whereas the 2H-MoS₂ was detected in high-loading MoS₂/hNCNC, consistent with the HRTEM observations (Fig. S5f). The electrochemical measurements reveal that the HER overpotential at 10 mA cm⁻² first decreases and then increases with the increase of MoS₂ content, reaching the lowest value for Pt-(20.8 wt%) MoS₂/hNCNC (Fig. S6a, b). The overpotential evolution of these Pt-MoS₂/hNCNC catalysts can be understood as follows. The low MoS₂ loading reduces the Pt-MoS₂ interaction while the high MoS₂ loading impedes the electron conductivity due to the increased layer number (Table

S1, Fig. S7), which are the main reasons for the deteriorated HER activities (Fig. S6a, b).

The Pt loading in Pt-(20.8 wt%) MoS₂/hNCNC was adjusted by changing the concentration of H₂PtCl₆ solution (Table S2). With increasing the Pt loading from 3.8 wt% through 6.1 wt% to 7.3 wt%, the HER overpotential decreases whereas the specific activity based on the Pt mass declines (Fig. S6c, d). By comprehensively considering both overpotential and mass activity, the Pt-MoS₂/hNCNC with 20.8 wt% MoS₂ and 3.8 wt% Pt is the optimal for studying the structure-performance relationship.

The morphological and structural characterizations of the optimal Pt-MoS₂/hNCNC and the Pt₁/hNCNC control are shown in Fig. 1. The HRTEM image shows that, except for a large number of bare surfaces of hNCNC, there are abundant vdW heterojunctions of MoS₂ monolayer and hNCNC in the optimal Pt-MoS₂/hNCNC (Fig. 1a). Such vdW heterostructures expose the (001) basal planes and edge active sites of MoS₂, and facilitate the electron transfer between MoS₂ and hNCNC (Table S1, Fig. S7). The high-angle annular dark-field scanning TEM (HAADF-STEM) images of the optimal Pt-MoS₂/hNCNC reveal that ~94% of bright spots on the “bare” surface area of hNCNC are smaller than 0.2 nm, indicating that Pt mainly exists as isolated atoms on hNCNC (Fig. 1b). On the MoS₂ area, Pt atoms mainly form island-like bright spot regions with the Pt-Pt distance greater than 4.5 Å (Fig. 1c), and the HRTEM and XRD results exclude the possibility of forming Pt clusters or nano-

particles (Fig. 1a, Fig. S5), indicating that Pt exists in the form of single-atom layer on MoS₂ monolayers. Thereafter, the optimal Pt-MoS₂/hNCNC is represented as Pt₁-MoS₂/hNCNC considering the single-atom dispersion of Pt.

The X-ray photoelectron spectroscopy (XPS) analysis of Pt₁-MoS₂/hNCNC verifies the presence of Pt, Mo, and S elements (Fig. 1d, Fig. S8). The Pt 4f spectrum of Pt₁/hNCNC reveals two sets of signals corresponding to Pt²⁺ and Pt⁴⁺ species [8]. For Pt₁-MoS₂/hNCNC, a pair of new signals corresponding to Pt-S species appears at 72.80 and 76.10 eV [16]. Compared with Pt-(38.7 wt%) MoS₂/hNCNC, the Pt 4f peaks for Pt₁-MoS₂/hNCNC shift toward the higher binding energy, while the S 2p peaks shift toward the lower binding energy, indicating the electron transfer from Pt to MoS₂ with strong Pt-support interactions (Fig. S8). In the X-ray absorption fine structure (XAFS) analysis (Fig. 1e), the normalized X-ray absorption near-edge structure (XANES) spectrum of Pt₁-MoS₂/hNCNC displays a stronger white-line intensity compared to Pt foil, signifying the dominated oxidation state of Pt. The white-line edge slightly shifts toward higher energy compared with Pt₁/hNCNC, and the shift becomes more significant with increasing MoS₂ loading (Fig. S9), which can be ascribed to the strong Pt-MoS₂ interaction. The *k*³-weighted extended XAFS (EXAFS) at the Pt L₃ edge reveals a dominant peak at 1.90 Å, corresponding to the Pt coordinated with light elements such as N, C, O, or S. Notably, no significant peak appears at 2.61 Å corresponding to the Pt-Pt bond (Fig. 1d)

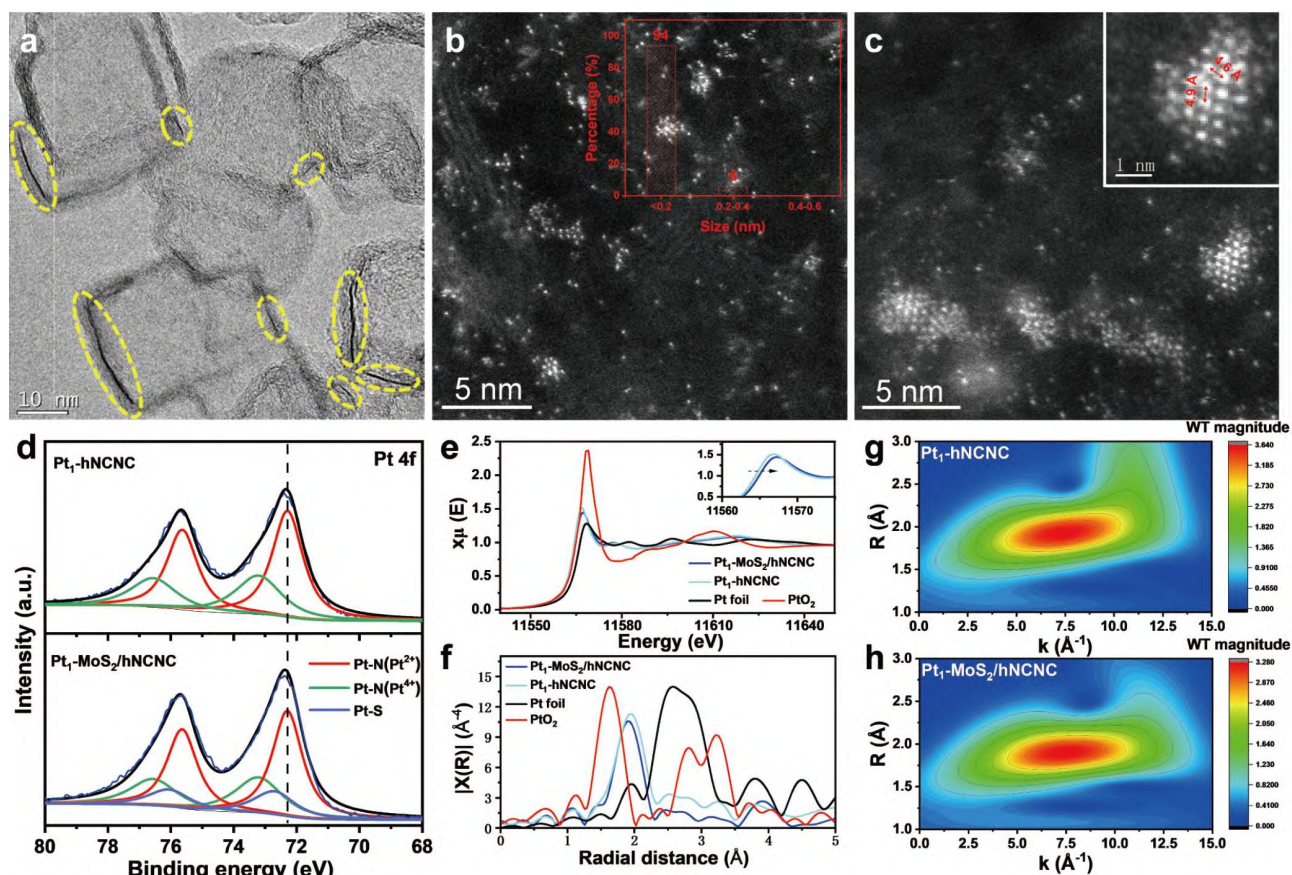


Figure 1 Morphology and structure of Pt₁-MoS₂/hNCNC and Pt₁/hNCNC. TEM image (a) and HAADF-STEM images (b, c) of Pt₁-MoS₂/hNCNC. Yellow dashed ellipses in (a) highlight the MoS₂ monolayers. Inset in (b) is corresponding size statistics of bright spots. Inset in (c) is an enlarged monolayer of Pt atoms. (d) Pt 4f XPS spectra. (e) Normalized XANES spectra at Pt L₃-edge, with Pt foil and PtO₂ for reference. Inset is the local enlargement. (f) *k*³-weighted *R*-space Fourier transformed spectra from EXAFS. (g, h) Wavelet transform spectra.

[26,27]. Similarly, wavelet transform analysis of Pt₁-hNCNC and Pt₁-MoS₂/hNCNC shows no prominent signal at 10.2 Å corresponding to the Pt-Pt coordination. This result confirms that Pt single atoms in Pt₁-MoS₂/hNCNC are coordinated with lighter elements, without the signal of clusters and nanoparticles (Fig. 1e, f, h). Fitting results (Fig. S10 and Table 3) show that the average bond lengths for Pt-S and Pt-N in Pt₁-MoS₂/hNCNC are 2.37 and 2.18 Å, respectively, without evident Pt-Pt bonds. These findings confirm that Pt exists as isolated single atoms in Pt₁-MoS₂/hNCNC, rather than clusters and nanoparticles, even though some Pt atoms form island-like single-atom layer (Fig. 1b, c).

Fig. 2 presents the HER performances of Pt₁-MoS₂/hNCNC, Pt₁-hNCNC and commercial Pt/C (20 wt%) in 0.5 M H₂SO₄ solution. The linear sweep voltammetry (LSV) polarization curves show that Pt₁-MoS₂/hNCNC achieves a current density of 10 mA cm⁻² at a low overpotential of 11 mV, which is obviously better than that of Pt/C (26 mV), Pt₁-hNCNC (16 mV) and reported Pt-based catalysts (Fig. 2a and Table S4). It is evident that the participation of MoS₂ monolayers promotes the HER activity of Pt₁-MoS₂/hNCNC to surpass that of Pt₁-hNCNC. The Tafel slope for Pt₁-MoS₂/hNCNC is 29 mV dec⁻¹, slightly lower than those for Pt₁-hNCNC and Pt/C, indicating its Volmer-Tafel mechanism for HER (Fig. 2b). At a potential of 20 mV, the Pt₁-MoS₂/hNCNC exhibits a mass activity of 5.6 A mg_{Pt}⁻¹, about 16 times higher than Pt/C (0.35 A mg_{Pt}⁻¹) and also larger than Pt₁-hNCNC (3.6 A mg_{Pt}⁻¹), showing the enhanced Pt utilization with the MoS₂/hNCNC heterostructure support (Fig. 2c). The turnover frequency (TOF) of Pt₁-MoS₂/hNCNC reaches 15.5 H₂ s⁻¹ at the potential of 50 mV, surpassing that of Pt₁-hNCNC (8.0 H₂ s⁻¹) and Pt/C (5.7 H₂ s⁻¹), showing the higher intrinsic activity of Pt₁-MoS₂/hNCNC for HER (Fig. S11). The Nyquist plots for Pt₁-MoS₂/hNCNC, Pt₁-hNCNC, and Pt/C display nearly semicircular shapes, showing a smaller charge

transfer resistance (R_{ct}) and better HER kinetics of Pt₁-MoS₂/hNCNC than those of Pt/C and Pt₁-hNCNC (Fig. 2d).

Moreover, the Pt₁-MoS₂/hNCNC demonstrates outstanding electrochemical stability for acidic HER. In a 0.5 M H₂SO₄ solution, the accelerated degradation tests (ADT) were performed using cyclic voltammetry (CV) at a scan rate of 100 mV s⁻¹, within a potential window of +0.2 to -0.2 V. After 5000 cycles, the overpotential for Pt₁-MoS₂/hNCNC at a current density of 50 mA cm⁻² is almost unchanged (39 mV), and decreases by only 1 mV after 10,000 cycles. In contrast, before and after 5000 and 10,000 CV scans, the corresponding overpotentials for Pt/C are 71, 73 and 76 mV, respectively, and those for Pt₁-hNCNC are 56, 59 and 62 mV. This result reveals the better HER stability of Pt₁-MoS₂/hNCNC than Pt₁/hNCNC and Pt/C (Fig. 2e). In addition, after a 100-hour chronoamperometry (CP) test at a current density of 50 mA cm⁻², only a slight potential drop of 2 mV is observed for Pt₁-MoS₂/hNCNC (Fig. 2f), accompanied by a minor aggregation of Pt species (Fig. S12). This result also corroborates the superb HER stability of Pt₁-MoS₂/hNCNC.

As demonstrated in our previous reports, the low-coordinated Pt-N₂ intermediate under HER working condition is the real active site for Pt₁/hNCNC, which enables the multi-H adsorption with an ideal ΔG_{H^*} of nearly 0 eV for the fourth H^{*} thereby the high activity [14]. The density functional theory (DFT) calculations were also conducted to understand the high HER activity and stability of the Pt single atoms anchored on MoS₂ as shown in Fig. 3. The optimized structure shows that the Pt atom is situated atop the Mo atom, forming the Pt-S₃ configuration. The Pt-S₃ site shows strong adsorption of H atom, with the ΔG_{H^*} of -0.04 and -0.91 eV for adsorbing the first and second H atoms, respectively (Fig. 3a). However, these two H^{*} atoms cannot combine to release a H₂ molecule. More H atoms are subsequently adsorbed on the Pt atom and the ΔG_{H^*} values for

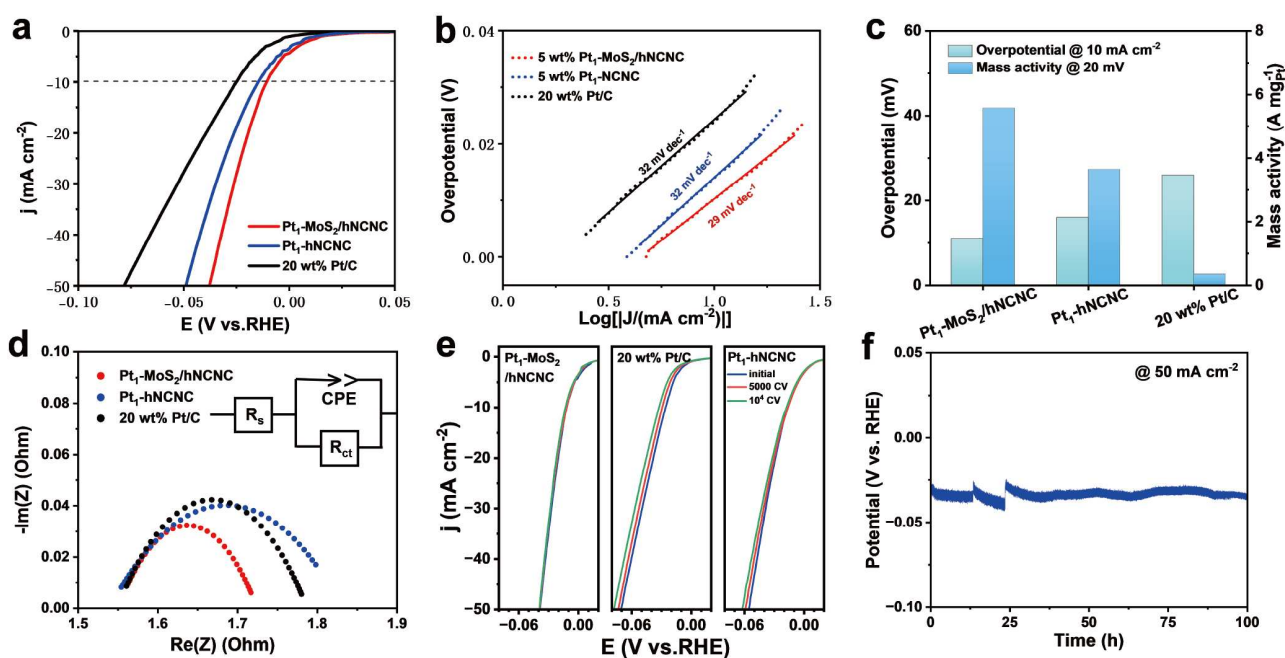


Figure 2 Electrocatalytic HER performances of Pt₁-MoS₂/hNCNC, Pt₁/hNCNC and commercial Pt/C. (a) Polarization curves. (b) Tafel slopes. (c) Overpotential at 10 mA cm⁻² and mass activities at 20 mV. (d) Nyquist plots. (e) Polarization curves before and after 5000 and 10,000 CV cycles. (f) CP curve at 50 mA cm⁻² of Pt₁-MoS₂/hNCNC.

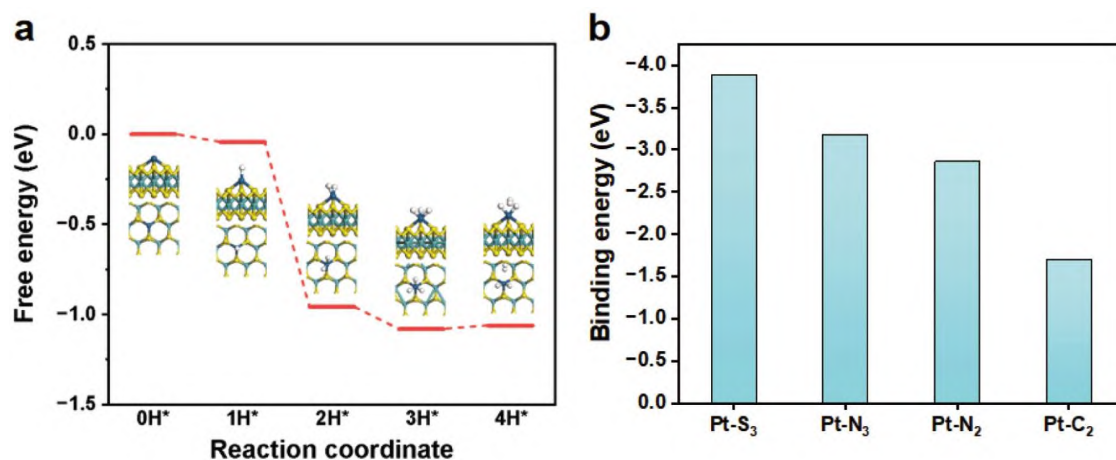


Figure 3 DFT results of HER performance. (a) Free energy diagram on Pt-S₃ sites during HER. (b) Binding energies of possible Pt single-sites in Pt₁-MoS₂/hNCNC.

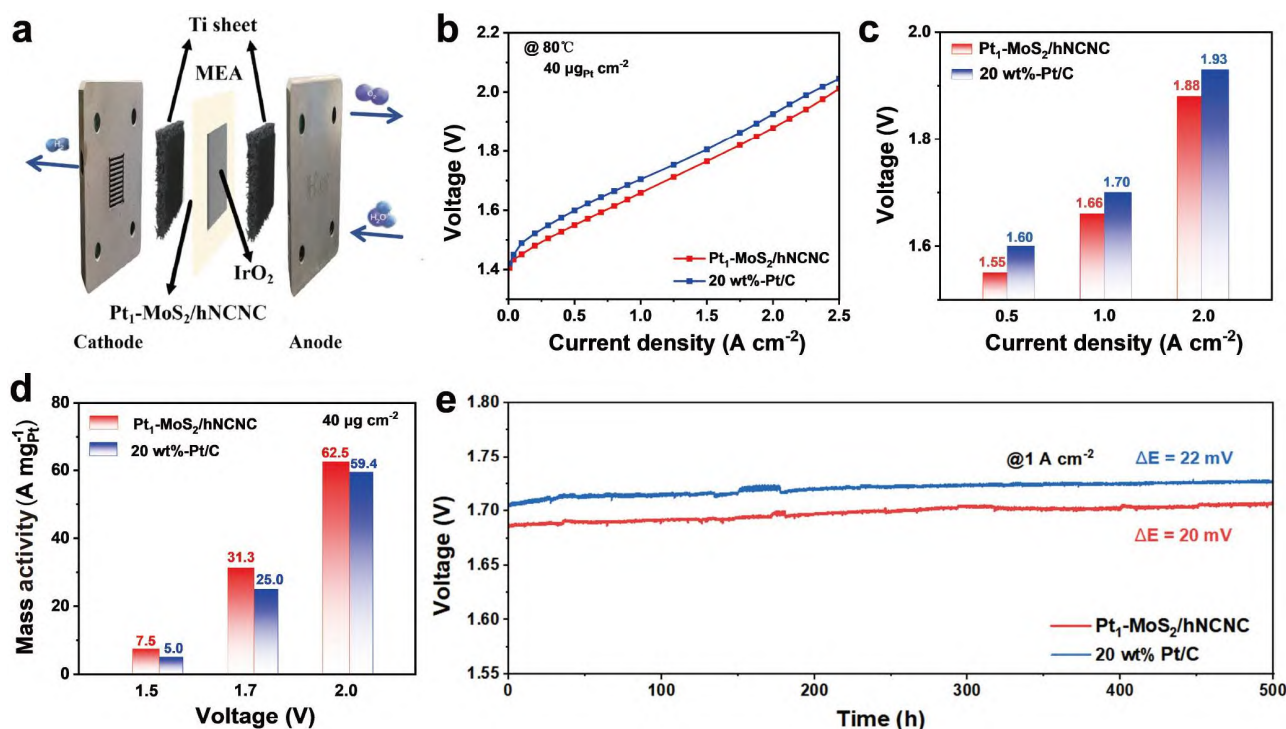


Figure 4 PEMWE performances of Pt₁-MoS₂/hNCNC and commercial 20 wt% Pt/C. (a) Schematic diagram of the device. (b) Polarization curves. (c) Cell voltages at different current densities. (d) Mass activity. (e) CP curves at 1.0 A cm⁻².

the third and fourth H atoms are -0.12 and 0.02 eV, respectively. Once the fourth H atom is adsorbed, a H₂ molecule can be desorbed, and then the HER cycles between Pt-S₃-2H* and Pt-S₃-4H* (Fig. 3a). It is learnt that the Pt atoms anchored to MoS₂ achieve HER through a multi-H adsorption mechanism, with the active site of Pt-S₃-2H*. The ΔG_{H*} for the third H adsorption is the descriptor of HER for Pt-MoS₂, and the value of -0.12 eV is very close to the ideal value of 0 eV, indicating its high intrinsic HER activity [15]. Therefore, for Pt₁-MoS₂/hNCNC, both Pt-S₃-2H* (on MoS₂) and Pt-N₂-3H* (on hNCNC) single-sites function as the key moieties for HER, contributing to its excellent HER performance.

The DFT calculations were performed to assess the stability of possible Pt single-atom sites in Pt₁-MoS₂/hNCNC (Fig. 3b). The binding energies of Pt atoms at Pt-S₃, Pt-N₃, Pt-N₂, and Pt-C₂ sites are -3.88, -3.18, -2.86 and -1.71 eV, respectively, indicating the stronger interaction between Pt atom and MoS₂ than that between Pt atom and hNCNC. Moreover, the energy and structural changes of Pt migration on MoS₂ in electrolyte are calculated, demonstrating a prohibitive 2.24 eV energy barrier (transition state) along the z direction and a high energy barrier of 1.0 eV in the xy plane at room temperature (Fig. S13). Accordingly, the Pt single-atoms anchored on the MoS₂ exhibit better stability than those on the hNCNC. The DFT results

elucidate the enhanced HER activity and stability of Pt₁-MoS₂/hNCNC on the vdW heterostructure support.

The HER performance of Pt₁-MoS₂/hNCNC was also evaluated in PEMWE as shown in Fig. 4. An MEA was assembled using Pt₁-MoS₂/hNCNC as the cathode catalyst and commercial IrO₂ as the anode catalyst (Fig. 4a). With an ultralow Pt loading of 40 μg_{Pt} cm⁻² (an order of magnitude lower than that in commercial PEMWE), the MEA achieves impressive current densities of 0.5 A cm⁻² at 1.55 V, 1.0 A cm⁻² at 1.66 V, and 2.0 A cm⁻² at 1.88 V, outperforming the MEA with the cathode catalyst of commercial 20 wt% Pt/C (Fig. 4b, c). Moreover, the Pt₁-MoS₂/hNCNC-based MEA demonstrates a superior mass activity of 31.3 A mg_{Pt}⁻¹ at 1.7 V, significantly exceeding the 25.0 A mg_{Pt}⁻¹ for the Pt/C-based MEA (Fig. 4d). The durability testing under a constant current density of 1 A cm⁻² shows that the Pt₁-MoS₂/hNCNC MEA maintains stable HER over 500 h, with a deterioration rate of 40 μV h⁻¹, obviously better than the 90 μV h⁻¹ for Pt₁/hNCNC and 44 μV h⁻¹ for Pt/C (Fig. 4e) [14]. Combining the advantages of ultralow Pt loading, industrial-level current density and high durability (Table S5), Pt₁-MoS₂/hNCNC MEAs present a high promise in the practical application of PEMWE.

CONCLUSIONS

In conclusion, this study demonstrates the successful development of highly efficient and durable Pt SAC for HER by utilizing a vdW heterostructure support composed of monolayer MoS₂ and hNCNC. The optimal Pt₁-MoS₂/hNCNC catalyst exhibits exceptional HER performance, achieving ultralow overpotentials (11 mV at 10 mA cm⁻²), high mass activity (5.6 A mg_{Pt}⁻¹ at -20 mV), and remarkable stability in acidic conditions. In the PEMWE, the MEA with an ultralow Pt loading of 40 μg_{Pt} cm⁻² delivers industrial-level current densities (1.0 A cm⁻² at 1.66 V) and superior durability (40 μV h⁻¹), outperforming commercial Pt/C and Pt₁-hNCNC. The enhanced performance is attributed to the synergistic effects of the vdW heterostructure, which provides optimized electronic structure, efficient charge/mass transport, and strong Pt-support interactions. This study demonstrates an innovative strategy to develop highly active and durable Pt SAC using vdW heterostructure supports. Future studies can be dedicated to developing more robust vdW heterostructure supports to further improve the HER performance of Pt SACs. Moreover, the unique vdW supports can also be used to construct advanced catalysts for various applications.

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- Yang ZX, Li XG, Yao QL, *et al.* 2022 roadmap on hydrogen energy from production to utilizations. *Rare Met*, 2022, 41: 3251–3267
- Bin S, Chen Z, Zhu Y, *et al.* High-pressure proton exchange membrane water electrolysis: current status and challenges in hydrogen production. *Int J Hydrogen Energy*, 2024, 67: 390–405
- Du Y, Liu J, Chen J, *et al.* Design principle and regulation strategy of noble metal-based materials for practical proton exchange membrane water electrolyzer. *Adv Energy Mater*, 2024, 15: 2404113
- Liu J, Liao J, Huang K, *et al.* A general strategy to remove metal aggregates toward metal–nitrogen–carbon catalysts with exclusive atomic dispersion. *Adv Mater*, 2023, 35: 2211398
- Makhsos A, Kandidayeni M, Pollet BG, *et al.* A perspective on increasing the efficiency of proton exchange membrane water electrolyzers—a review. *Int J Hydrogen Energy*, 2023, 48: 15341–15370

- Fang S, Zhu X, Liu X, *et al.* Uncovering near-free platinum single-atom dynamics during electrochemical hydrogen evolution reaction. *Nat Commun*, 2020, 11: 1029
- Jin H, Ha M, Kim MG, *et al.* Engineering Pt coordination environment with atomically dispersed transition metal sites toward superior hydrogen evolution. *Adv Energy Mater*, 2023, 13: 2204213
- Zhang Z, Chen Y, Zhou L, *et al.* The simplest construction of single-site catalysts by the synergism of micropore trapping and nitrogen anchoring. *Nat Commun*, 2019, 10: 1657
- Zhang Z, Zhu Y, Asakura H, *et al.* Thermally stable single atom Pt/m-Al₂O₃ for selective hydrogenation and CO oxidation. *Nat Commun*, 2017, 8: 16100
- Ren Y, Tang Y, Zhang L, *et al.* Unraveling the coordination structure–performance relationship in Pt₁/Fe₂O₃ single-atom catalyst. *Nat Commun*, 2019, 10: 4500
- Zhao D, Chen Z, Yang W, *et al.* MXene (Ti₃C₂) vacancy-confined single-atom catalyst for efficient functionalization of CO₂. *J Am Chem Soc*, 2019, 141: 4086–4093
- Han B, Guo Y, Huang Y, *et al.* Strong metal–support interactions between Pt single atoms and TiO₂. *Angew Chem Int Ed*, 2020, 59: 11824–11829
- Li F, Han GF, Bu Y, *et al.* Unveiling the critical role of active site interaction in single atom catalyst towards hydrogen evolution catalysis. *Nano Energy*, 2022, 93: 106819
- Tian J, Xia M, Cheng X, *et al.* Understanding Pt active sites on nitrogen-doped carbon nanocages for industrial hydrogen evolution with ultralow Pt usage. *J Am Chem Soc*, 2024, 146: 33640–33650
- Cheng X, Mao C, Tian J, *et al.* Correlation between heteroatom coordination and hydrogen evolution for single-site Pt on carbon-based nanocages. *Angew Chem Int Ed*, 2024, 63: e202401304
- Shi Z, Zhang X, Lin X, *et al.* Phase-dependent growth of Pt on MoS₂ for highly efficient H₂ evolution. *Nature*, 2023, 621: 300–305
- Zhou Y, Liu Y, Zhang M, *et al.* Rationally designed hierarchical N, P co-doped carbon connected 1T/2H-MoS₂ heterostructures with cooperative effect as ultrafast and durable anode materials for efficient sodium storage. *Chem Eng J*, 2022, 433: 133778
- Wang H, Xiao X, Liu S, *et al.* Structural and electronic optimization of MoS₂ edges for hydrogen evolution. *J Am Chem Soc*, 2019, 141: 18578–18584
- Liu Y, Huang Y, Duan X. Van der Waals integration before and beyond two-dimensional materials. *Nature*, 2019, 567: 323–333
- Regan EC, Wang D, Paik EY, *et al.* Emerging exciton physics in transition metal dichalcogenide heterobilayers. *Nat Rev Mater*, 2022, 7: 778–795
- Liu Y, Weiss NO, Duan X, *et al.* Van der Waals heterostructures and devices. *Nat Rev Mater*, 2016, 1: 16042
- Wu Q, Yang L, Wang X, *et al.* Mesostructured carbon-based nanocages: an advanced platform for energy chemistry. *Sci China Chem*, 2020, 63: 665–681
- Wu Q, Yang L, Wang X, *et al.* Carbon-based nanocages: a new platform for advanced energy storage and conversion. *Adv Mater*, 2020, 32: 1904177
- Wu Q, Yang L, Wang X, *et al.* From carbon-based nanotubes to nanocages for advanced energy conversion and storage. *Acc Chem Res*, 2017, 50: 435–444
- Xu F, Feng B, Shen Z, *et al.* Oxygen-bridged Cu binuclear sites for efficient electrocatalytic CO₂ reduction to ethanol at ultralow overpotential. *J Am Chem Soc*, 2024, 146: 9365–9374
- Wang C, Wang K, Feng Y, *et al.* Co and Pt dual-single-atoms with oxygen-coordinated Co–O–Pt dimer sites for ultrahigh photocatalytic hydrogen evolution efficiency. *Adv Mater*, 2021, 33: 2003327
- Mosallanezhad A, Wei C, Ahmadian Koudakan P, *et al.* Interfacial synergies between single-atomic Pt and CoS for enhancing hydrogen evolution reaction catalysis. *Appl Catal B-Environ*, 2022, 315: 121534

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Conflict of interest The authors declare that they have no conflict of interest.

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范德华异质结载体锚定的Pt单原子催化剂用于耐久性析氢反应

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摘要 单原子催化剂的活性和稳定性与载体的结构密切相关. 本文采用范德华异质结载体构建了具有高析氢(HER)活性和耐久性的Pt单原子催化剂. 这种独特的载体由附着在分级结构氮掺杂碳纳米笼(hNCNC)上的单层MoS₂组成, Pt在hNCNC上以单原子形式存在, 在MoS₂上以岛状单原子层形式存在. 优化后的Pt₁-MoS₂/hNCNC催化剂在0.5 M H₂SO₄溶液中表现出低过电位(11 mV@10 mA cm⁻²)和高质量活性 (5.6 A mg_{Pt}⁻¹@-20 mV), 优于商业Pt/C. 相对于商业Pt/C, Pt₁-MoS₂/hNCNC在质子交换膜水电解槽中表现出更优的长效稳定性. Pt₁-MoS₂/hNCNC的优异HER性能归因于其可调的电子结构、Pt与MoS₂/hNCNC的强相互作用及更快的电荷转移动力学. 本项研究展示了利用范德华异质结载体构建高活性和长耐久性Pt单原子催化剂的新策略.